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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 3	
TITLE : Method Overloading and Static Members	

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Questions:

1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.
2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

Programme

1 :

```
J Demo.java
1  class Calculator {
2
3      static int count = 0;
4
5      // Addition of integers
6      int add(int a, int b) {
7          count++;
8          return a + b;
9      }
10
11     // Addition of decimals
12     double add(double a, double b) {
13         count++;
14         return a + b;
15     }
16
17     public static void main(String[] args) {
18
19         Calculator c = new Calculator();
20
21         System.out.println("Integer Addition: " + c.add(10, 20));
22
23         System.out.println("Decimal Addition: " + c.add(10.5, 20.5));
24
25         System.out.println("Total Calculations: " + count);
26     }
27 }
```

2 :

```
1  class Restaurant {
2
3      static int totalOrders = 0;
4
5      // Dine-in
6      double calculateBill(double foodAmount, double tax) {
7
8          totalOrders++;
9
10         return foodAmount + tax;
11     }
12
13     // Takeaway
14     double calculateBill(double foodAmount, double tax, double packingCharge) {
15
16         totalOrders++;
17
18         return foodAmount + tax + packingCharge;
19     }
20 }
```

```

21 // Delivery
22 double calculateBill(double foodAmount, double tax,
23     double packingCharge, double deliveryCharge) {
24     totalOrders++;
25     return foodAmount + tax + packingCharge + deliveryCharge;
26 }
27
28
29
30 public static void main(String[] args) {
31     Restaurant r = new Restaurant();
32
33     System.out.println("Dine-in Bill: ₹" +
34         r.calculateBill(500, 50));
35
36     System.out.println("Takeaway Bill: ₹" +
37         r.calculateBill(500, 50, 30));
38
39     System.out.println("Delivery Bill: ₹" +
40         r.calculateBill(500, 50, 30, 50));
41
42     System.out.println("Total Orders: " + totalOrders);
43 }
44
45

```

Output

1 :

```

● piyushrohokale@Piyushs-MacBook-Air assignment06 % javac Demo.java && java Demo
Data written successfully.

Employee Details:
Employee ID: 518
Employee Name: Piyush
Department: Finance
○ piyushrohokale@Piyushs-MacBook-Air assignment06 %

```

2 :

```
● piyushrohokale@Piyushs-MacBook-Air assignment06 % javac Demo.java && java Demo
Data written successfully.

Employee Details:
Employee ID: 518
Employee Name: Piyush
Department: Finance%
○ piyushrohokale@Piyushs-MacBook-Air assignment06 %
```

Github:

Java_assignments / assignment03 /			Add file ▾	...
VyankateshRohokale A3			adb5008 · 5 days ago	History
Name	Last commit message	Last commit date		
..				
.DS_Store	A3	5 days ago		
Addition.class	A3	5 days ago		
Addition.java	A3	5 days ago		
Assignment 3.pages	A3	5 days ago		

Conclusion

The program show the concept of method overloading using the addition functions differ from there parameters. It also demonstrates the use of a static variable and a static method , which belong to the class and can be accessed without creating an object.

